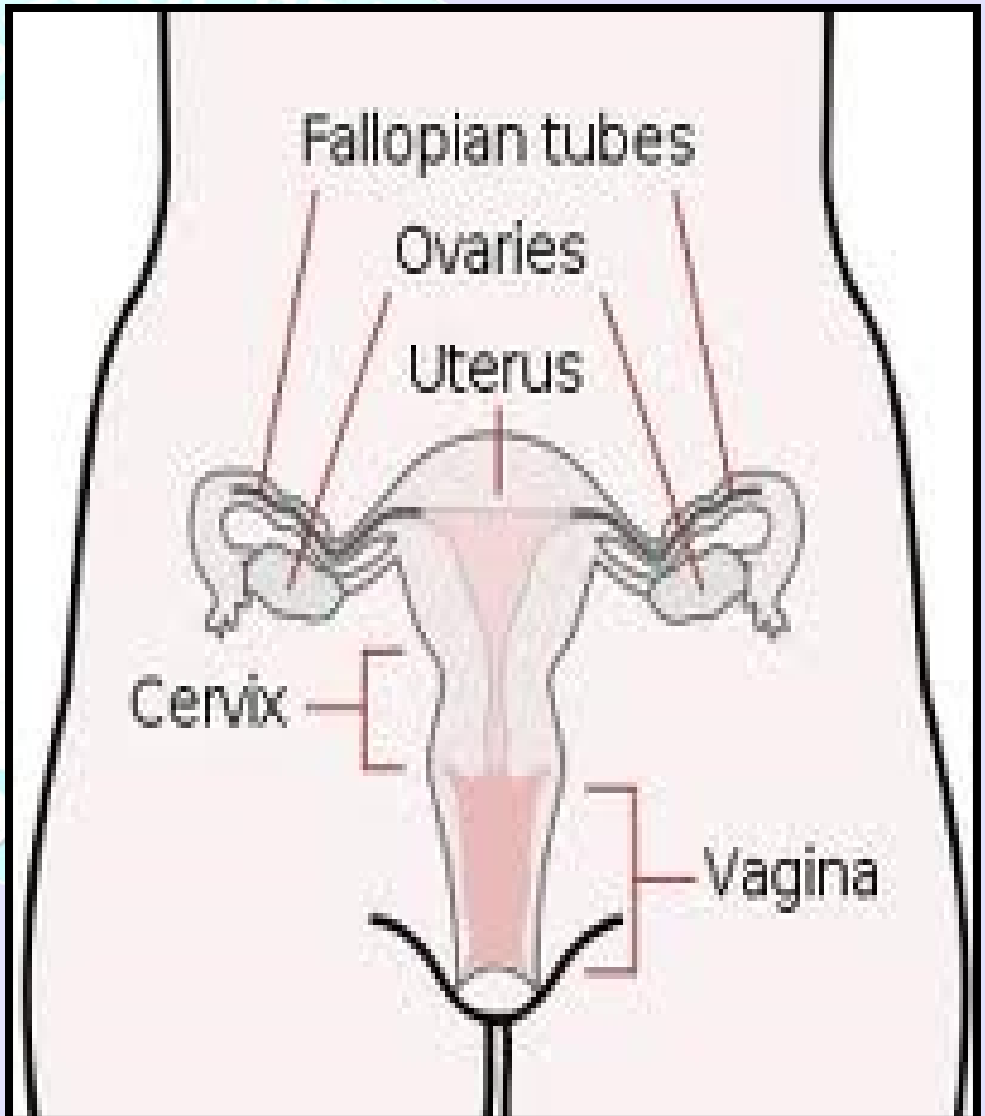
Female Reproductive organs

LEARNING OBJECTIVES

- At the end of the lecture, student should be able to
  - Discuss the brief gross anatomy of each part of female genital system
  - Recognize the different parts of female genital tract and their location in pelvic cavity
  - State the blood supply, nerve supply and lymphatic drainage
  - Mention the microscopic features of each part in relation to ovulation, menstruation and pregnancy

Parts of Female Reproductive Organs

- Uterus
- Fallopian Tubes
- Ovaries
- Mammary Glands



UTERUS

Uterus The uterus (Latin [L.], womb) is a thick-walled, pear-shaped muscular organ

Dimension

averaging 7 to 8 cm in length,

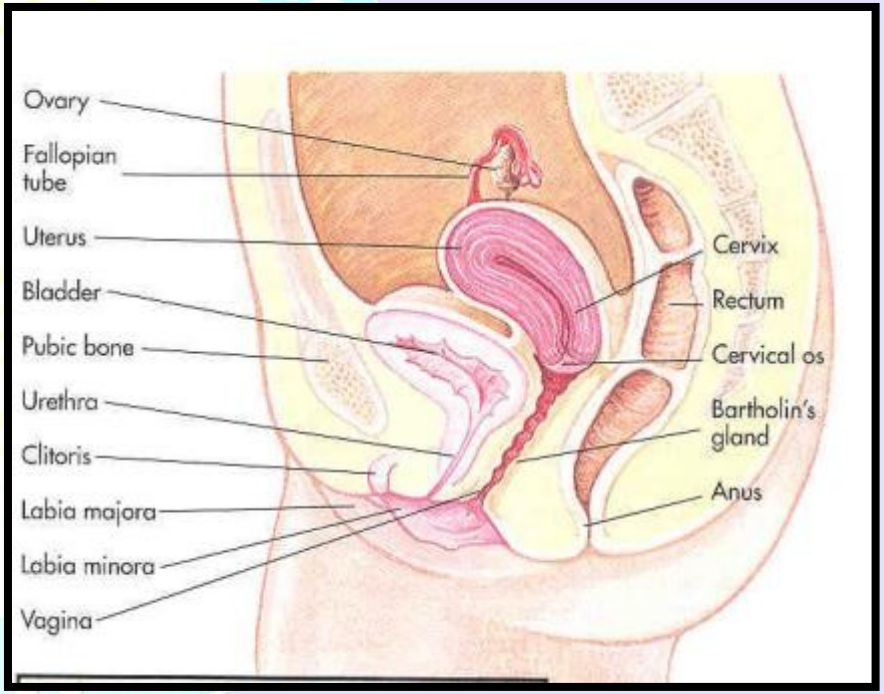
5 to 7 cm in width at its superior part,

2 to 3 cm in wall thickness.

The uterus consists of two major parts

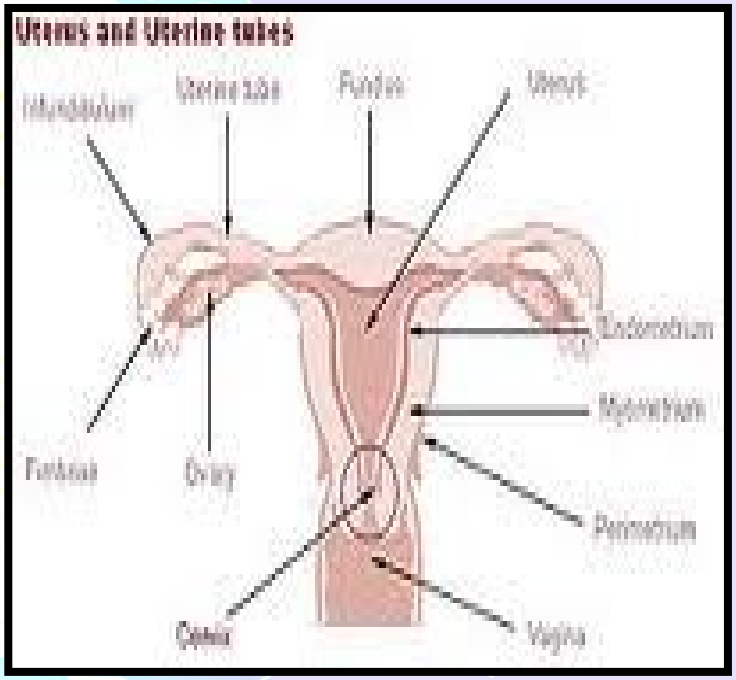
Body, the expanded superior two thirds

Cervix, the cylindrical inferior one third



Body of the uterus

The **body of the uterus** narrows from the **fundus,** the rounded, superior part of the body, to the **isthmus**, the 1-cm-long constricted region between the body and cervix (L., neck).

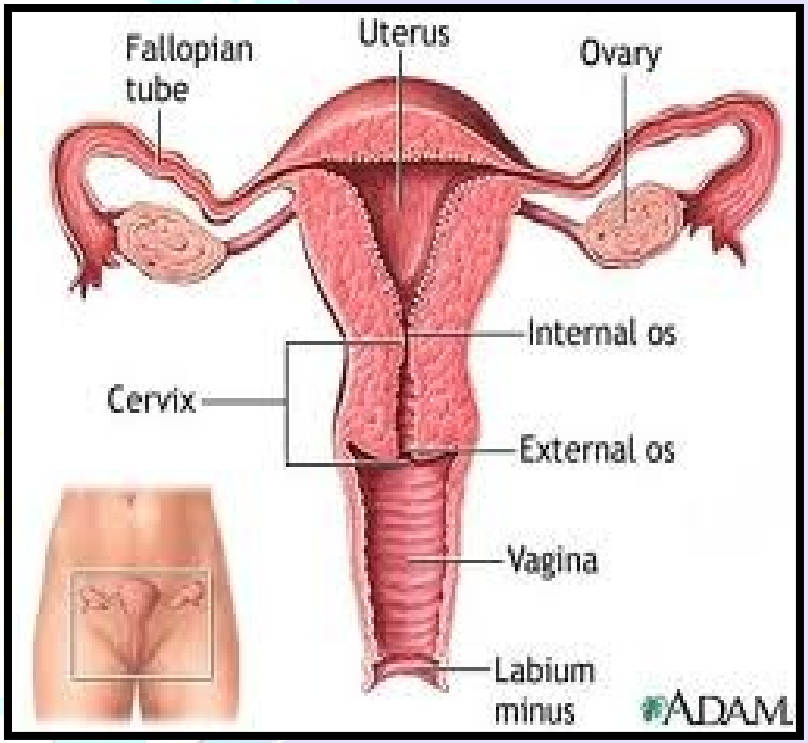


Cervix of the uterus

The **cervix of the uterus** is its tapered vaginal end that is nearly cylindrical in shape. The lumen of the cervix, the **cervical canal**, has a constricted opening at each end.

The **internal os** communicates with the cavity of the uterine body

The **external os** communicates with the vagina.



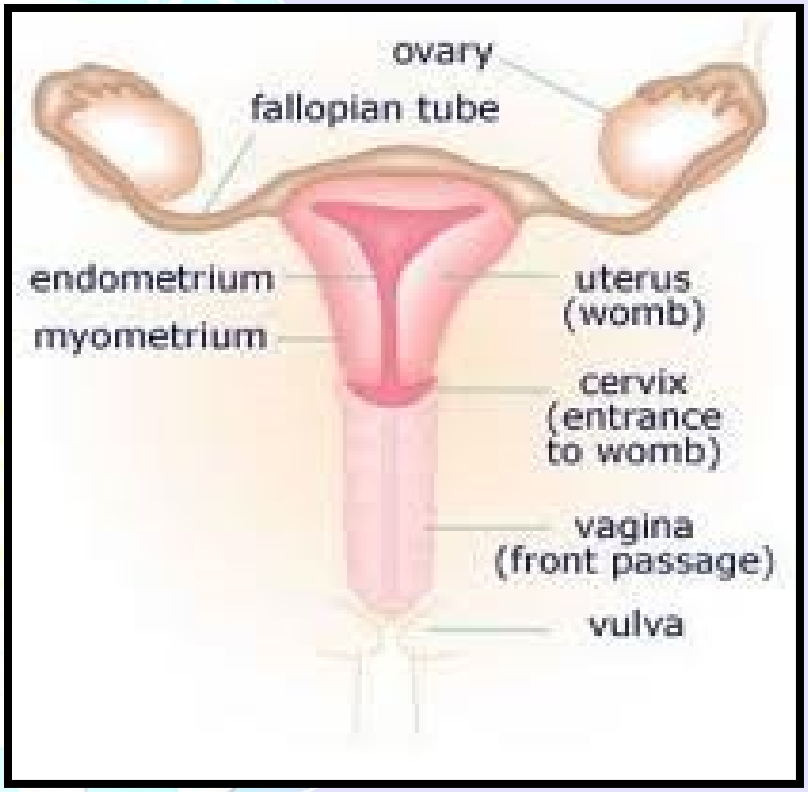
Layers of Uterus

The walls of the body of the uterus consist of three layers:

Perimetrium, the thin external layer

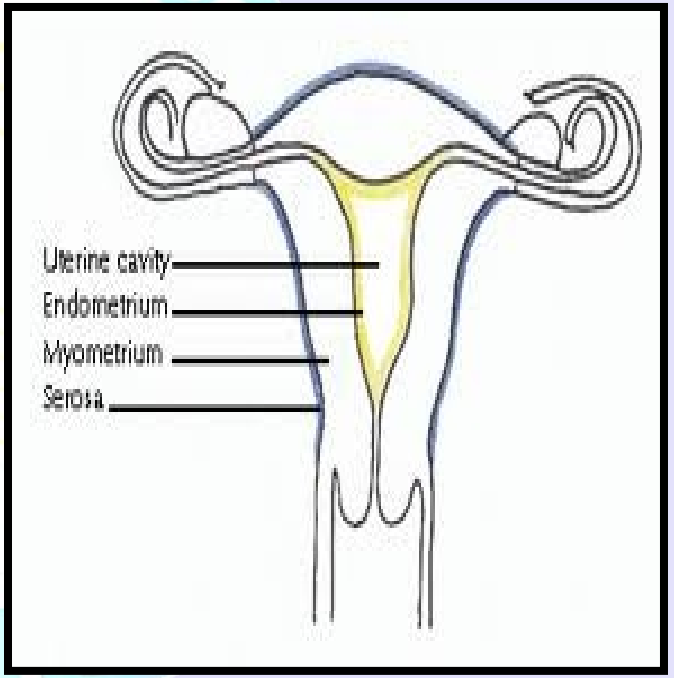
Myometrium, the thick smooth muscle layer

Endometrium, the thin internal layer



Perimetrium

The **perimetrium** is a peritoneal layer that is firmly attached to the myometrium.



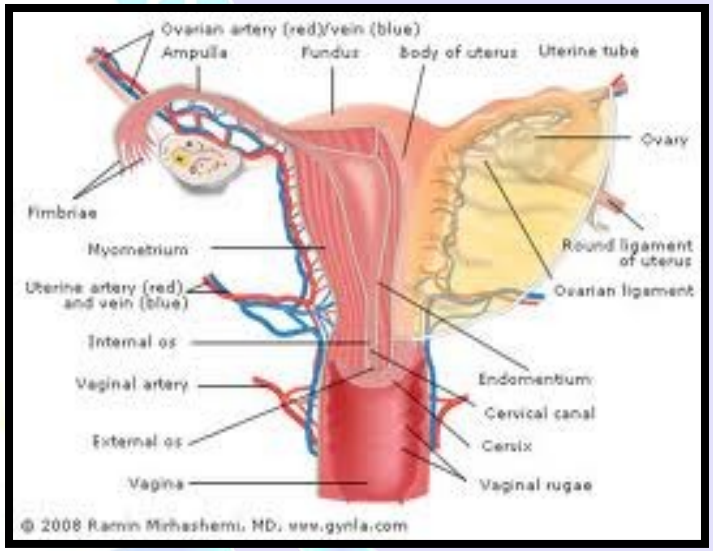
Endometrium

During the luteal (secretory) phase of the menstrual cycle, three layers of the endometrium can be distinguished microscopically

A thin, **compact layer** consisting of densely packed, connective tissue around the necks of the uterine glands

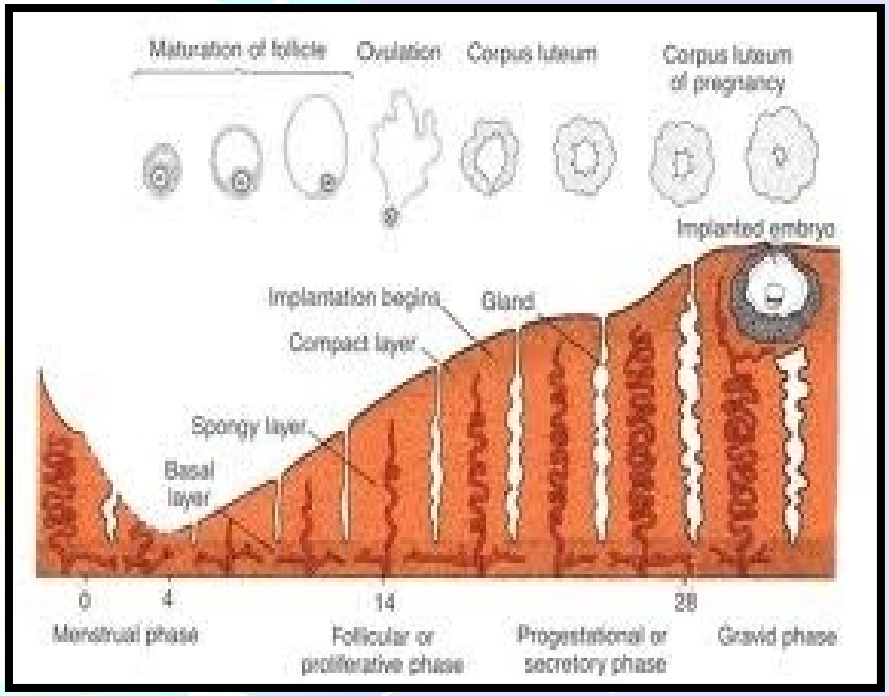
A thick, **spongy layer** composed of edematous connective tissue containing the dilated, tortuous bodies of the uterine glands

A thin, **basal layer** containing the blind ends of the uterine glands



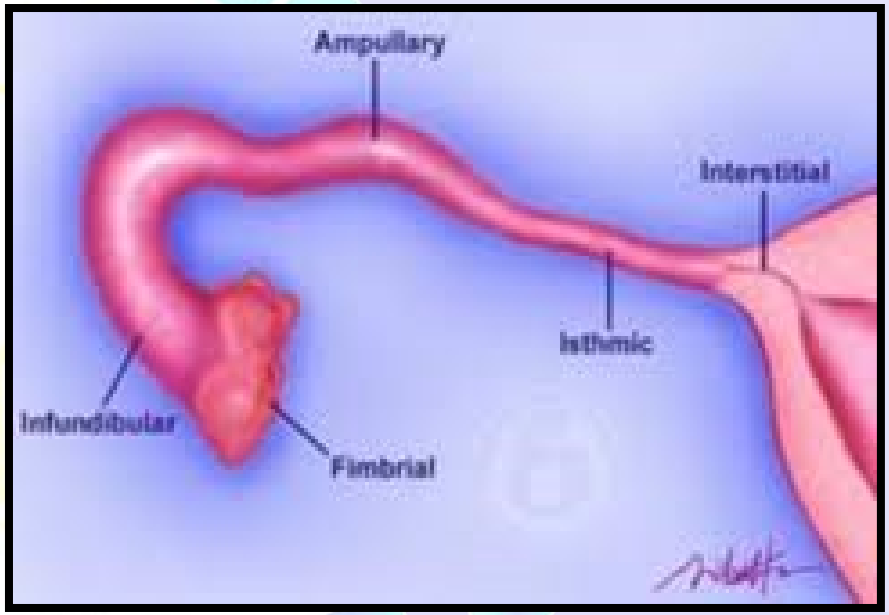
Endometrium

At the peak of its development, the endometrium is 4 to 5 mm thick. The basal layer of the endometrium has its own blood supply and is not sloughed off during menstruation. The compact and spongy layers, known collectively as the **functional layer**, disintegrate and are shed during menstruation and after parturition (delivery of a baby).



Uterine Tubes

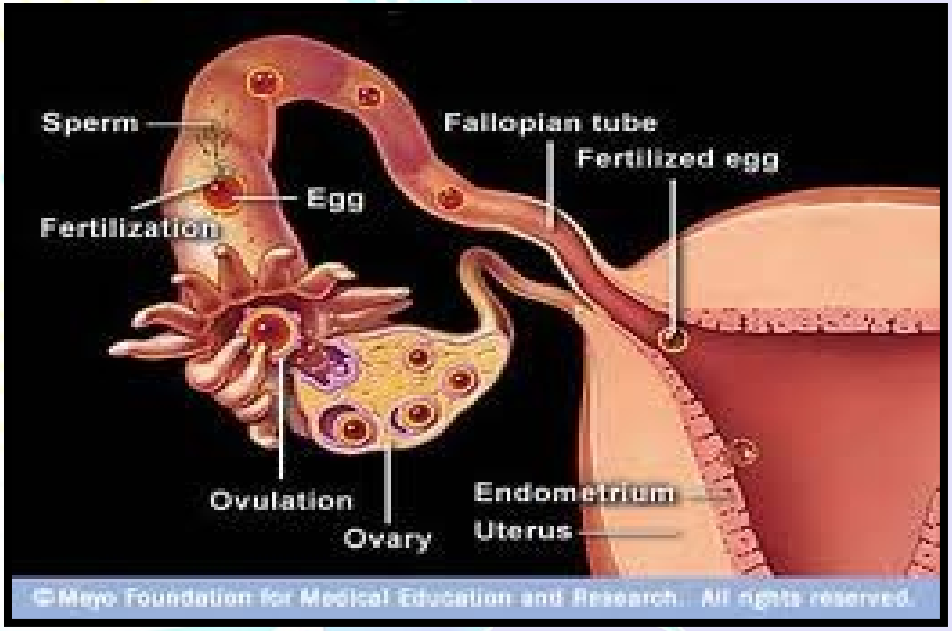
The uterine tubes, approximately 10 cm long and 1 cm in diameter, extend laterally from the **horns** (L., *cornua*) **of the uterus**. Each tube opens at its proximal end into the horn of the uterus and into the peritoneal cavity at its distal end.



Parts of Fallopian Tube

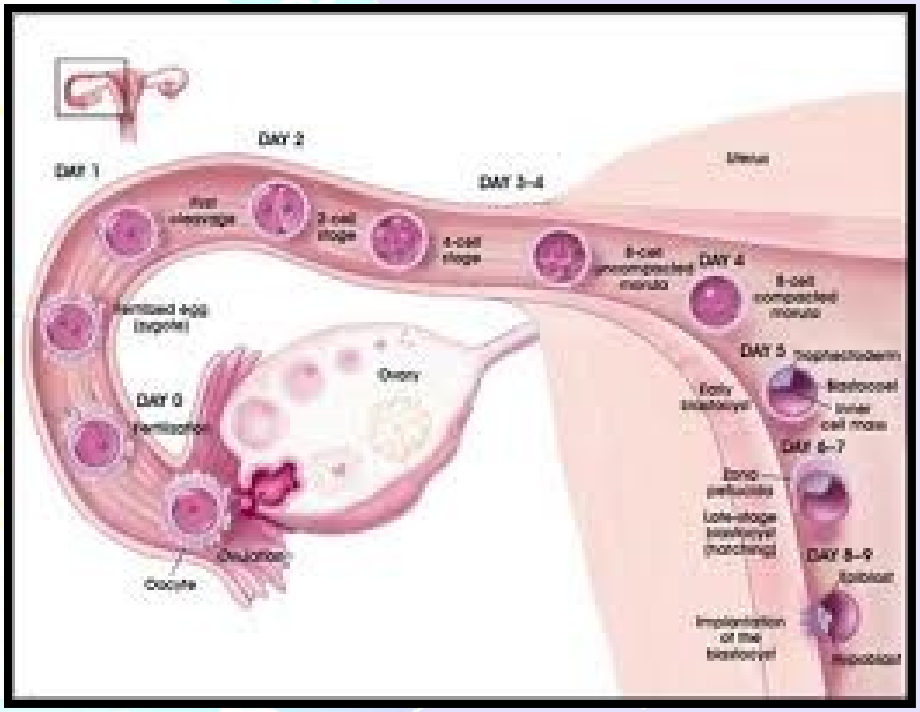
For descriptive purposes, *the uterine tube is divided into four parts*:

- the infundibulum
- the ampulla
- the isthmus
- the uterine part.



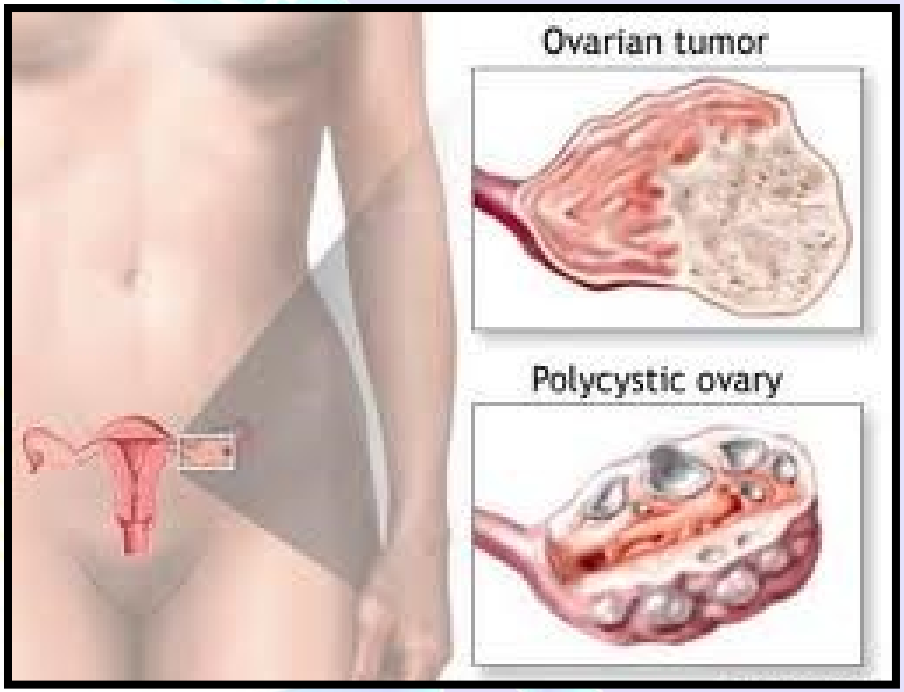
Function of Fallopian tube

The tubes carry oocytes from the ovaries and sperms entering from the uterus to reach the fertilization site in the **ampulla of the uterine tube**. The uterine tube also conveys the cleaving zygote to the uterine cavity.



Ovaries

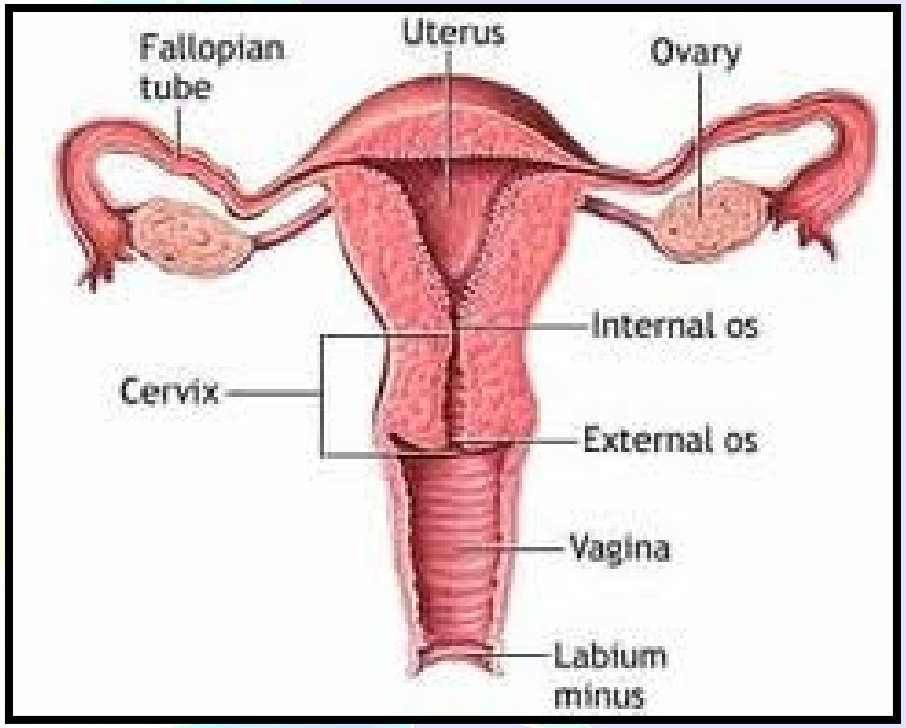
- The ovaries are almond-shaped reproductive glands located close to the lateral pelvic walls on each side of the uterus that produce oocytes. The ovaries also produce estrogen and progesterone, the hormones responsible for the development of secondary sex characteristics and regulation of pregnancy.



Vagina

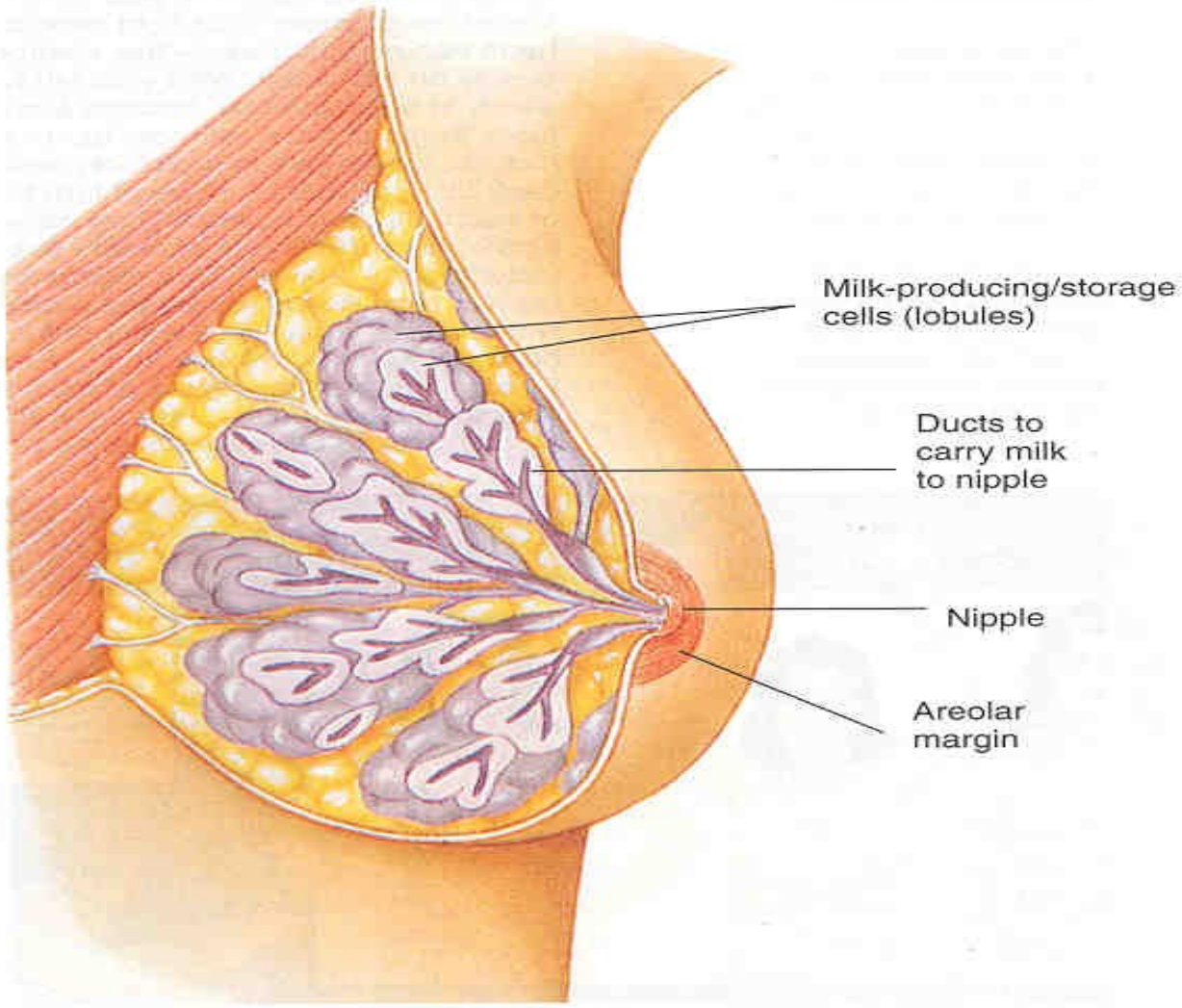
The **vagina** is a fibro muscular tubular tract leading from the uterus to the exterior of the body in female mammals

The vagina is the place where semen from the anatomic male is deposited into the anatomically female person.



BREASTS

- Organs of sexual arousal
- Contain mammary glands
- Consist of connective tissue that serves as support
- Each breast contain 15-25 clusters called lobes
- Each lobule is connected by ducts that open into the nipples
- The nipples are made up of erectile tissue
- The pigmented around the nipples are called the areola
- Breast size is determined primarily by heredity
- Size also depends on the existing fat and glandular tissue
- Breasts may exhibit cyclical changes, including increased swelling and tenderness prior to menstruation
- Benign breast changes refer to fibrocystic disease
- Lumps or masses that are noncancerous


Milk-producing/storage cells (lobules)
Ducts to carry milk to nipple
Nipple
Areolar margin

REFERENCES

- **Keith L. Moore Developing Human**

  8th Edition

  Chapter-2

  pages 22 -23

THANK YOU